MIRZO ULUG‘BEK NOMIDAGI O‘ZBEKISTON MILLIY UNIVERSITETI
Amaliy matematika va kompyuter tahlili kafedrasi

MATLATIPOV SANATBEK

O‘zbek tili gaplarini neyron tarmoqlar asosida sintaktik tahlil qilishning matematik modeli

05.01.07 – Matematik modellashtirish. Sonli usullar va dasturlar majmui

Kafedra ilmiy seminari

Ilmiy rahbar: f.-m.f.d., professor Aripov Mersaid Mirsidikovich

Toshkent – 2026

Tadqiqot obyekti: korpusdagi yozuv va model chiqishi — sintaktik daraxt

«Bu bo‘limdagi asarlar bizga bu xazinaning eshigini ochib beradi.» (UzUDT korpusi, s658-gap)

Korpus: CoNLL-U yozuvi (qo‘lda belgilangan)

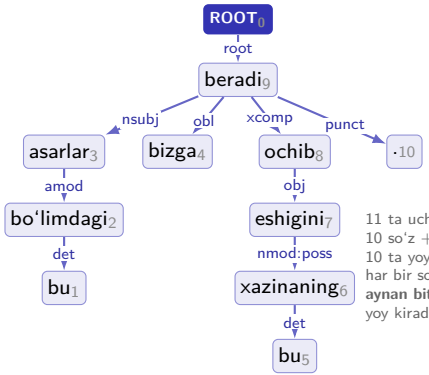
ID	FORM	LEMMA	UPOS	HEAD	DEPREL
1	bu	bu	PRON	2	det
2	bo‘limdagi	bo‘lim	ADJ	3	amod
3	asarlar	asar	NOUN	9	nsubj
4	bizga	biz	PRON	9	obl
5	bu	bu	PRON	6	det
6	xazinaning	xazina	NOUN	7	nmod:poss
7	eshigini	eshik	NOUN	8	obj
8	ochib	och	VERB	9	xcomp
9	beradi	ber	VERB	0	root
10	.	.	PUNCT	9	punct

\+ XPOS va morfologik belgilar, masalan eshigini: Case=Acc, Person[psor]=3.
HEAD — so‘zning **o‘zagi** raqami, **DEPREL** — bog‘lanish turi.

model

neyron tarmoq + MST

Chiqish: yo‘naltirilgan daraxt (graf)



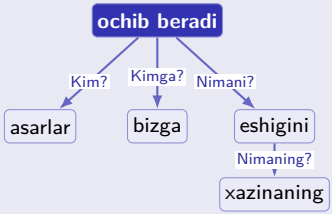
11 ta uch:
10 so‘z + ildiz;
10 ta yoy;
har bir so‘zga **aynan bitta** yoy kiradi

684	7 582	$\kappa \geq 0{,}91$
gap	token	6 annotator

Kirish: so‘zlar ketma-ketligi $x = (w_1, \ldots, w_n) \implies$ **Chiqish:** har bir so‘zning o‘zagi h_i (HEAD) va bog‘lanish turi r_i (DEPREL), ya’ni **yo‘naltirilgan daraxt**. **Vazifa:** shu daraxtni avtomatik topish.

Sintaktik daraxt nima uchun kerak: gapning ma'nosini ochish

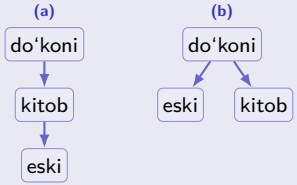
1. Kim? Kimga? Nimani?



Predikat va uning argumentlari:
OCHIB BERMOQ(*kim*: asarlar; *kimga*: biz; *nimani*: xazina eshigi).

2. Bir xil so'zlar — turli daraxt

«eski kitob do'koni»



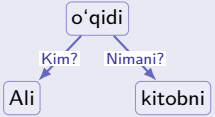
eski **kitoblar** sotiladigan do'kon

do'konning o'zi eski

⇒ **turli ma'no**

3. Turli tartib — bir xil daraxt

«Ali kitobni o'qidi.» ≡ *«Kitobni Ali o'qidi.»*



⇒ **bir xil ma'no**: o'zbek tilida so'z tartibi erkin, ma'noni qo'shimchalar va daraxt belgilaydi.

Xulosa

Gapning ma'nosi so'zlar ro'yxatida emas, ular orasidagi **graf tuzilmasida**. To'g'ri daraxt — **semantik tahlilning asosi**: mashina tarjimasi (SOV → SVO tartib), savol-javob tizimlari, matndan axborot ajratish, grammatik tekshiruv.

Mavzuning dolzarbligi

O‘zbek tili — 38 mln dan ortiq kishi so‘zlashadigan **agglyutinativ** turkiy til. Gapning **sintaktik tuzilishi** mashina tarjimasi, axborot qidiruvi va savol-javob tizimlari uchun asos.

Kam resursli muhit: Universal Dependencies (UD) tizimida o‘zbek tili uchun bor-yo‘g‘i **1 184 ta gap** sintaktik annotatsiyalangan.

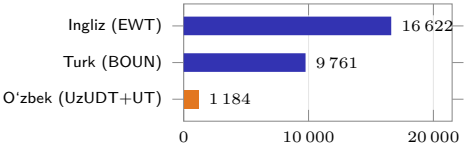
Kelishik, egalik, son, zamon va shaxs ma’nolari **qo‘shimchalar zanjiri** bilan ifodalanadi ⇒ so‘z shakllari ko‘p, lug‘at siyrak.

Kichik tanlanmada ishonchli ishlaydigan, **matematik asoslangan** model va **sonli usullar** (aniq dekodlash, turg‘un o‘qitish) zarur.

So‘z shakli: morfemalar va BPE qism so‘zlari

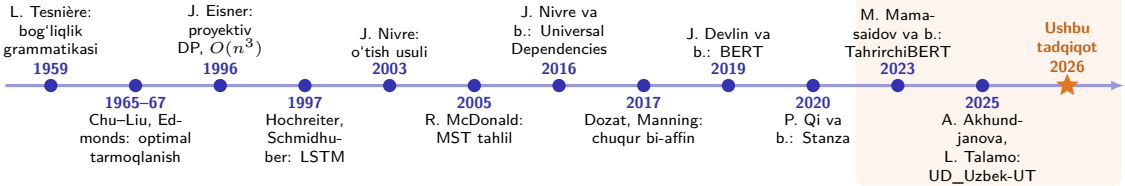


UD korpuslari hajmi (gaplar soni)



Manba: universaldependencies.org. BPE bo‘linishi — TahrirchiBERT tokenizatorining haqiqiy natijasi.

Muammoning o‘rganilganlik darajasi



O‘zbekiston olimlari

M. M. Aripov, M. X. Xakimov, G‘. R. Matlatipov, N. Abdurakhmonova, M. S. Sharipov, E. R. Quriyozov, B. B. Elov, U. I. Salaev, M. Mamasaidov

Xorijiy va MDH olimlari

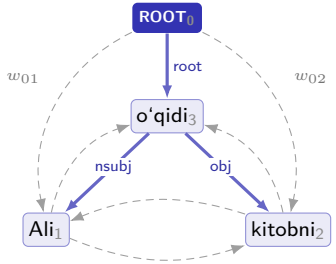
L. Tesnière, J. Edmonds, R. Tarjan, J. Eisner, J. Nivre, R. McDonald, T. Dozat, C. D. Manning, P. Qi, J. Devlin, M.-C. de Marneffe, D. Zeman, M. Straka, G. Eryiğit, U. A. Tukeyev (Qozog‘iston)

Hal etilmagan masala

O‘zbek tili uchun (i) UD sintaktik korpuslari juda kichik va heterogen; (ii) BPE qism so‘zlaridan **agglyutinativ so‘z vektorini** hosil qilish operatori asoslanmagan; (iii) aniq (MST) va ochko‘z dekodlash hamda modelning **turg‘unligi** tizimli o‘rganilmagan.

Masalaning qo‘yilishi: gapning graf modeli

«Ali kitobni o‘qidi» gapining grafi G_x



$n = 3$: $|A| = n^2 = 9$ ta yoy (shtrix), qalin — bitta qamrovchi daraxt; jami $|\mathcal{T}(x)| = 3^2 = 9$ ta daraxt.

Yoy $j \to i$: j — **o‘zak uch**, i — **tobe uch**; har bir yoyga vazn w_{ji} beriladi.

Ta’rif 1 (gap grafi va bog‘liqlik daraxti)

$x = (w_1, \ldots, w_n)$ — gap; $G_x = (V, A, W)$ — vaznli to‘liq yo‘naltirilgan graf: $V = \{0, 1, \ldots, n\}$ (0 — ildiz), $A = \{j \to i : \ i \geq 1,\ j \neq i\}$, $|A| = n^2$; $W = (w_{ji})$ — vazn matritsasi.

Bog‘liqlik daraxti $y \subset A$ — 0 ildizli **yo‘naltirilgan qamrovchi daraxt** (arborestsensiya): har bir $i \geq 1$ uchun kirish darajasi $\deg^-(i) = 1$ (yagona **o‘zak** h_i), sikl yo‘q, $\deg^+(0) = 1$; yoy belgisi $r_i \in \mathcal{R}$, $|\mathcal{R}| = 39$.

Lemma 1 (A. Cayley formulasi asosida)

$|\mathcal{T}(x)| = n^{n-1}$: $n = 10 \Rightarrow 10^9$; $\bar{n} = 11{,}4 \Rightarrow \approx 10^{11}$. *($n \leq 7$ da barcha daraxtlarni sanab tekshirdim.)*

Masala (P): eng og‘ir qamrovchi daraxtni topish

$\hat{y} = \arg\max_{y \in \mathcal{T}(x)} W(y)$, $W(y) = \sum_{(j \to i) \in y} w_{ji}$,
$w_{ji} = \log P_\theta(h_i = j \mid x)$.
Sifat: $\mathrm{UAS} = \frac{1}{N} \sum_i \mathbf{1}[\hat{h}_i = h_i]$, $\mathrm{LAS} = \frac{1}{N} \sum_i \mathbf{1}[\hat{h}_i = h_i \wedge \hat{r}_i = r_i]$.

Tadqiqotning maqsadi, vazifalari, obyekti va predmeti

Tadqiqotning maqsadi

Kam resursli sharoitda o‘zbek tili gaplarini avtomatik morfosintaktik tahlil qilishning neyron tarmoqli **matematik modelini**, uni o‘qitish va dekodlashning **sonli usullarini** ishlab chiqish hamda ularni **dasturiy majmua** sifatida amalga oshirish.

Tadqiqotning vazifalari

1. badiiy va ta’limiy matnlar asosida oltin standart UD korpusi (**UzUDT**) yaratish va uning graf xossalarini tahlil qilish;
2. tahlil masalasini **graflar nazariyasi** tilida — vaznli to‘liq yo‘naltirilgan grafda maksimal vaznli qamrovchi daraxtni topish masalasi sifatida qo‘yish;
3. Stanza graf asosidagi tahlilchisini o‘zbek tili uchun **o‘zgartirish**: TahrirchiBERT kanali va qism so‘zlarni birlashtirish operatori Π;
4. daraxtni topish (Chu–Liu–Edmonds) va o‘qitish (Adam) **sonli usullarini** qo‘llash, vaqt va aniqlik ko‘rsatkichlarini tadqiq qilish;
5. aniqlik va **turg‘unlikni** tajribada baholash; dasturiy majmuani yaratish.

Obyekt va predmet

Obyekt: o‘zbek tilidagi matnlar va ularning UD sintaktik korpuslari (UzUDT, UT).

Predmet: sintaktik tahlilning graf-ehtimollik modeli, uni o‘qitish va dekodlashning sonli usullari, dasturiy majmuasi.

Usullar

Graflar nazariyasi (qamrovchi daraxtlar), kombinatorika, stoxastik optimallashtirish, ehtimollar nazariyasi, matematik statistika, neyron tarmoqlar.

Tadqiqotning ilmiy yangiligi

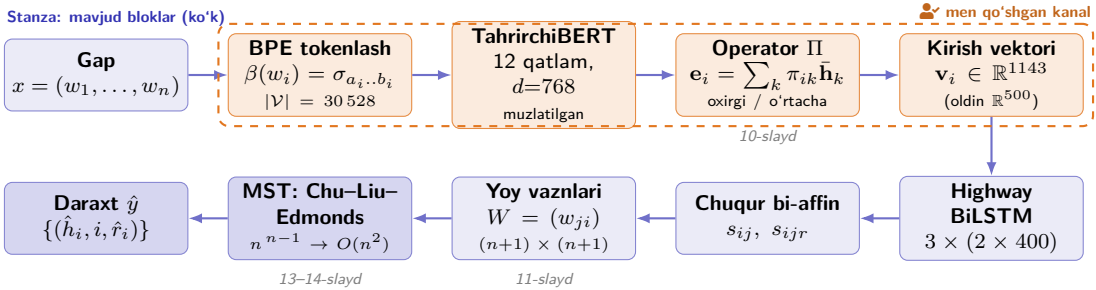
1. O‘zbek tilining badiiy va ta’limiy matnlari uchun **ilk bor** UD standartidagi oltin standart sintaktik korpus — **UzUDT** yaratilgan (684 gap, 7 582 token, 6 annotator, $\kappa = 0{,}91$–$0{,}95$) va UD v2.17 tarkibida ochiq e’lon qilingan; u hozirda eng katta ochiq o‘zbek UD korpusi. [2, 16-sl.]

2. Stanza kutubxonasining graf asosidagi bi-affin tahlilchisi o‘zbek tili uchun **o‘zgartirilgan**: kirish qatlamiga TahrirchiBERT kanali qator-stoxastik Π operatori orqali qo‘shilgan ($\mathbb{R}^{500} \to \mathbb{R}^{1143}$); o‘zbek tili uchun **ilk bor** «oxirgi qism so‘z» operatorining o‘rtacha qiymatdan ustunligi statistik tasdiqlangan ($+2{,}6$ LAS, $p \approx 0{,}002$). [9–11, 17-sl.]

3. Tahlil masalasi vaznli to‘liq yo‘naltirilgan grafda **maksimal vaznli qamrovchi daraxtni** topish masalasi sifatida qo‘yilgan va o‘zgartirilgan modelda Chu–Liu–Edmonds algoritmi bilan aniq yechilgan: izlash murakkabligi n^{n-1} variantdan $O(n^2)$ amalga tushirilgan (11 so‘zli gapda $\approx 1{,}7$ sutka $\to$ 0,42 ms); o‘zbek tili uchun **ilk bor** graf va o‘tish usullari bir xil test to‘plamida taqqoslangan. [6, 12–14, 18-sl.]

4. O‘zbek tili uchun **ilk bor** ikki UD korpusi (UzUDT+UT) birlashtirilib, korpuslararo empirik risk Adam$\to$AMSGrad usuli bilan minimallashtirilgan va yaqinlashish tahlil qilingan; tahlilchining orfografik g‘alayonlarga **turg‘unligi** miqdoriy baholangan; $\approx$3 100 qatorli **dasturiy majmua** yaratilgan. [15, 18, 19-sl.]

Men o‘zgartirgan model: Stanza graf asosidagi bi-affin tahlilchisi



	Stanza (E1.1)	O‘zgartirilgan (E2.1)
Kirish vektori $\mathbf{v}_i$	$\mathbb{R}^{500}$ (FastText)	$\mathbb{R}^{1143}$ (+ $\mathbf{e}_i \in \mathbb{R}^{768}$)
Qism so‘z → so‘z (Π)	—	oxirgi / o‘rtacha
O‘qitiladigan parametrlar	24,5 mln	27,6 mln*
Validatsiya: LAS / UPOS, %	55,28 / 79,19	**58,23** / **83,54**

* + TahrirchiBERT ning 109,5 mln parametri (muzlatilgan, o‘qitilmaydi).

$n^{n-1} \rightarrow O(n^2)$

daraxtni izlash murakkabligi (vaqt bo‘yicha)

+2,95 LAS, +4,35 UPOS

BERT kanali hisobiga (validatsiya)

Men qo‘shgan kanal: qism so‘z vektorlarini so‘zga o‘tkazish operatori Π

Ta’rif 3 (BPE va qatlamlar)

$\beta:\ w_i \mapsto (\sigma_{a_i}, \dots, \sigma_{b_i}),\ \ m_i = b_i - a_i + 1$ — qism so‘zlar soni.
Kontekstli tasvir (Stanza sxemasi):
$\bar{\mathbf{h}}_k = \frac{1}{4}\sum_{\ell=9}^{11} \mathbf{H}_k^{(\ell)} \in \mathbb{R}^{768}$.

Ta’rif 4 (birlashtirish operatori)

$\mathbf{E} = \Pi\,\bar{\mathbf{H}},\ \ \mathbf{e}_i = \sum_{k=1}^{M} \pi_{ik}\bar{\mathbf{h}}_k,\ \ \Pi \in \mathbb{R}^{n\times M}$ — **qator-stoxastik** matritsa:

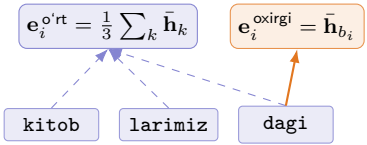
$$\pi_{ik}^{\text{oxirgi}} = \mathbf{1}[k = b_i], \qquad \pi_{ik}^{\text{o‘rt}} = \frac{\mathbf{1}[a_i \le k \le b_i]}{m_i}.$$

Gipoteza (suffiks lokalizatsiyasi)

O‘zbek tilida grammatik ko‘rsatkichlar so‘zning **o‘ng chetida** $\Rightarrow \bar{\mathbf{h}}_{b_i}$ kelishik va shaxs-son ma’lumotini to‘playdi; o‘rtacha qiymat bu signal ulushini $1/m_i$ gacha kamaytiradi.

TahrirchiBERT tokenizatorining haqiqiy natijasi

kitoblarimizdagi	kitob	larimiz	dagi		
o‘quvchilarimizdagi	o	‘	quvchi	larimiz	dagi
maktablarimizning	maktab	larimizning			
ko‘rganmisiz	ko	‘	rgan	misiz	
xazinaning	xazina	ning			



o‘/g‘ dagi tutuq belgisi (‘) alohida qism so‘zga ajraladi.

Yoy vaznlari: chuqur bi-affin baholash va ehtimollik modeli

Ta'rif 5 (chuqur bi-affin baholovchi)

$\mathbf{a}_i^d = \mathrm{ReLU}(\mathbf{W}_d\mathbf{r}_i + \mathbf{b}_d)$, $\mathbf{a}_j^h = \mathrm{ReLU}(\mathbf{W}_h\mathbf{r}_j + \mathbf{b}_h) \in \mathbb{R}^{400}$ (**tobe**, **o'zak**);

$s_{ij} = [\mathbf{a}_i^d; 1]^\top \mathbf{U} [\mathbf{a}_j^h; 1]$, $\mathbf{U} \in \mathbb{R}^{401\times 401}$; yoy belgisi: s_{ijr}.

Lemma 2 (bi-affin forma — bilinear forma)

$\mathbf{U} = \begin{bmatrix} \mathbf{U}_{11} & \mathbf{u}_{12} \\ \mathbf{u}_{21}^\top & u_{22} \end{bmatrix}$ bo'lsa, $s_{ij} = \mathbf{a}_i^{d\top}\mathbf{U}_{11}\mathbf{a}_j^h + \mathbf{u}_{21}^\top\mathbf{a}_j^h + c_i$,

$c_i = \mathbf{a}_i^{d\top}\mathbf{u}_{12} + u_{22}$.

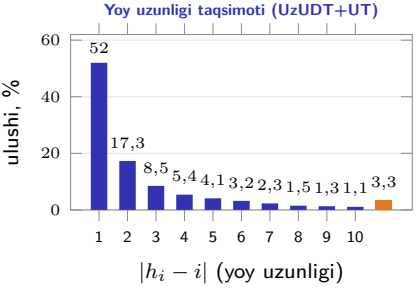
softmax siljishga invariant $\Rightarrow$ c_i ta'sir qilmaydi; $\mathbf{u}_{21}^\top\mathbf{a}_j^h$ — j ning **o'zak bo'lish** aprior bahosi.

Vazn matritsasi W (yoy yo'nalishi va Koshi yadrosi bilan)

$\tilde{s}_{ij} = s_{ij} + \log\sigma\big(\lambda_{ij}\,\mathrm{sgn}(j-i)\big) - \log\big(1 + \frac{1}{2}(|j-i| - \delta_{ij})^2\big)$;

$w_{ji} = \log \mathrm{softmax}_j(\tilde{s}_{ij}) = \log P_\theta(h_i = j \mid x)$,

$P_\theta(r_i = r \mid x, h_i = j) = \mathrm{softmax}_r(s_{ijr})$.



84,6 %
o'zak uch o'ngda (SOV)

$\bar{d} = 2{,}72$
o'rtacha yoy uzunligi

Tinish belgilari va ildiz yoylarisiz, 13 512 yoy. Og'ir dumli taqsimot yoy uzunligi uchun Koshi tipidagi yadro tanlovini asoslaydi.

Sonli usullar: ikki xil optimallashtirish masalasi

	(P1) Daraxtni topish (dekodlash)	(P2) Parametrlarni topish (o‘qitish)
Noma’lum	daraxt $y \in \mathcal{T}(x)$	parametrlar $\theta \in \mathbb{R}^p$, $p = 27{,}6 \cdot 10^6$
Maqsad funksiyasi	$W(y) = \sum_{(j \to i) \in y} w_{ji} \;\to\; \max$	$\widehat{R}(\theta)$ — o‘rtacha yo‘qotish $\to$ min
Masala turi	diskret: $\lvert\mathcal{T}(x)\rvert = n^{\,n-1}$ variant	uzluksiz, qavariq emas, stoxastik
Sonli usul	Chu–Liu–Edmonds (MST)	Adam $\to$ AMSGrad, erta to‘xtatish
Nima bo‘yicha yutuq	**vaqt**: $n^{\,n-1}$ variant $\to O(n^2)$ amal	**aniqlik**: LAS $24{,}7\,\% \to 58{,}2\,\%$
Erishilgan natija	11 so‘zli gap: $\approx 1{,}7$ sutka $\to$ 0,42 ms	E2.1: 1 000 qadam, 6,3 daqiqa (GPU)



❓ **«Optimallashtirish nima bo‘yicha?»** — (P1) da **hisoblash vaqti** bo‘yicha (eksponensial $\to$ kvadratik), (P2) da **yo‘qotish funksiyasi**, ya’ni aniqlik bo‘yicha.

Sonli usul (1): maksimal vaznli qamrovchi daraxt — Chu–Liu–Edmonds algoritmi

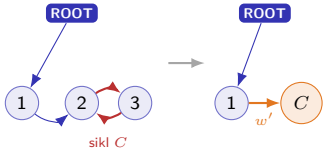
Teorema 1 (Chu–Liu–Edmonds) va algoritm

G_x vaznli to‘liq yo‘naltirilgan graf, $w_{ji} = \log P_\theta(h_i = j \mid x)$ bo‘lsin. Model yoylar bo‘yicha faktorlanganda (P) ning yechimi — **maksimal vaznli yo‘naltirilgan qamrovchi daraxt**; uni Chu–Liu–Edmonds algoritmi **aniq** topadi:

1) har bir i uchun eng og‘ir kiruvchi yoy: $h_i = \arg\max_j w_{ji}$; **2)** sikl bo‘lmasa — yechim optimal; **3)** sikl C bitta uchga qisqartiriladi, $w'_{j\to C} = \max_{i\in C}\left(w_{ji} - w_{h_i i}\right)$, masala qayta yechiladi; **4)** $\deg^+(0) > 1$ bo‘lsa, nomzod ildizlar bo‘yicha qayta yechiladi.

Men bajargan ishlar

1. Tahlilni maksimal vaznli qamrovchi daraxt masalasiga keltirdim.
2. O‘zgartirgan modelimda daraxtni shu algoritm bilan **aniq** topdim.
3. Tekshirdim: $n \le 7$ da 1 560 masalada to‘liq saralash bilan **100 %** mos.

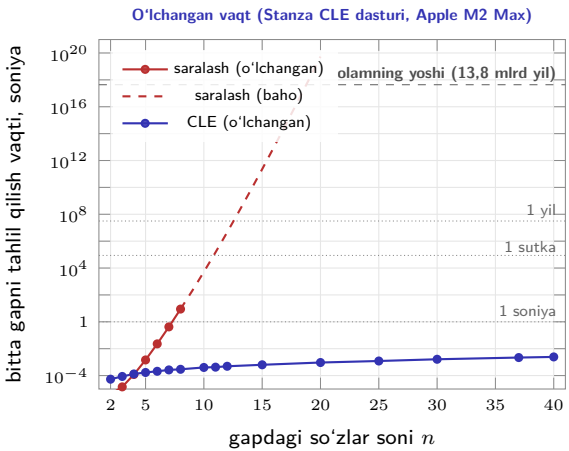


Murakkablik — vaqt bo‘yicha

To‘liq saralash: n^{n-1} daraxt $\times\, O(n) = O(n^n)$ — **eksponensial**.
CLE: bitta iteratsiya $O(n^2)$ (W ning har bir elementi bir marta ko‘riladi), iteratsiyalar o‘rtacha 1–2 $\Rightarrow O(n^2)$.

9,2 % gaplar noproyektiv (109 / 1 184): proyektiv usullar ($O(n^3)$ dinamik dasturlash) ularni ifodalay olmaydi, MST ifodalaydi.

Hisoblash tajribasi: n^{n-1} dan $O(n^2)$ ga — vaqt bo‘yicha yutuq



n	n^{n-1}	To‘liq saralash	CLE
5	625	1,5 ms	0,17 ms
8	$2{,}1 \cdot 10^{6}$	8,9 s	0,29 ms
11	$2{,}6 \cdot 10^{10}$	$\approx$1,7 sutka*	**0,42 ms**
20	$5{,}2 \cdot 10^{24}$	$\approx 1{,}7 \cdot 10^{12}$ yil*	0,93 ms
37	$2{,}9 \cdot 10^{56}$	$\approx 1{,}7 \cdot 10^{44}$ yil*	2,2 ms

* o‘lchangan tezlik bo‘yicha baho (0,52 mks / variant); 11 — o‘rtacha, 37 — eng uzun test gapi. Vaznlar tasodifiy (sikllar ko‘p — og‘ir holat).

$\approx 3{,}5 \cdot 10^{8}$ **marta tez**
o‘rtacha (11 so‘zli) gapda: 1,7 sutka $\rightarrow$ 0,42 ms

Tajriba natijalari

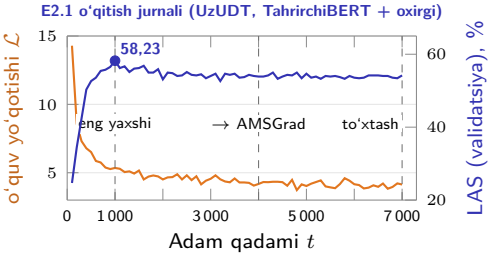
1. 19 so‘zli gapni saralash **olamning yoshidan** uzoq.
2. Test to‘plami (198 gap): CLE — jami **0,09 s**.
3. $n = 20 \ldots 150$: $t_{\mathrm{CLE}} \propto n^{1{,}56}$ (n^2 dan sekin).

Sonli usul (2): parametrlarni baholash — Adam usuli bilan o‘qitish

Masala va Adam iteratsiyasi

$\hat{\theta} = \arg\min_{\theta \in \mathbb{R}^p} \widehat{R}(\theta)$, $p = 27{,}6$ mln;
$\widehat{R} = \lambda \widehat{R}_{\text{UzUDT}} + (1-\lambda)\widehat{R}_{\text{UT}}$, $\lambda = \frac{451}{781} \approx 0{,}58$ ($\mathcal{L}$ — bir so‘zga o‘rtacha manfiy log-ehtimollik).

$\mathbf{g}_t = \nabla_\theta \mathcal{L}_{B_t}(\theta_{t-1})$, $\|\mathbf{g}_t\|_2 \le 1$;
$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t$, $\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\mathbf{g}_t^2$;
$\theta_t = \theta_{t-1} - \alpha\, \hat{\mathbf{m}}_t / \left(\sqrt{\hat{\mathbf{v}}_t} + \varepsilon\right)$, $\hat{\mathbf{m}}_t = \mathbf{m}_t/(1-\beta_1^t)$, $\hat{\mathbf{v}}_t = \mathbf{v}_t/(1-\beta_2^t)$;
$\alpha = 3 \cdot 10^{-3}$, $\beta_1 = 0{,}9$, $\beta_2 = 0{,}95$, $\varepsilon = 10^{-6}$; partiya $\le 5\,000$ so‘z (UzUDT: 1 davr = 2 qadam).



Men bajargan ishlar

1. Ikki korpus uchun maqsad funksiyasini tuzdim.
2. BERT ni muzlatdim: $137 \to 27{,}6$ mln parametr (≈5 marta kam).
3. To‘xtatish: 3 000 qadam yaxshilanmasa — AMSGrad, yana 3 000 — to‘xtash.
4. Jurnallashtirib, 5 tajribada yaqinlashishni tahlil qildim.

Natija (E2.1)

Yo‘qotish $14{,}29 \to 5{,}35$ ($-63\,\%$), LAS $24{,}7 \to \mathbf{58{,}2}\,\%$ — **1 000 qadamda**, 375 s (RTX A6000). Keyin LAS $\approx 54\,\%$ ga tushadi (haddan tashqari moslashish) $\Rightarrow$ eng yaxshi nuqta saqlanadi. 5 tajribada eng yaxshi qadam: 800–1 300.

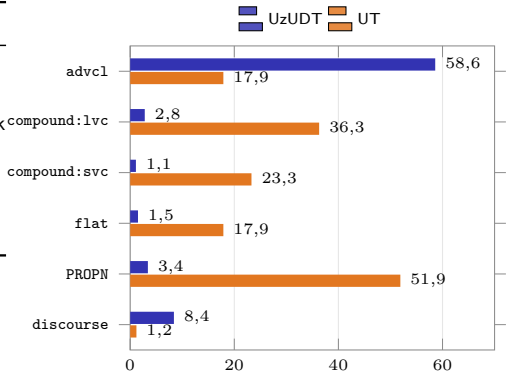
Ma’lumotlar va tajriba dizayni

	UzUDT (men yaratgan)	**UT**
Janr	badiiy, ta’limiy	yangilik, badiiy
Gap / token	684 / 7 582	500 / 5 930
Annotatsiya	qo‘lda, 6 annotator	yarim avtomatik
Kelishuv (κ, %)	lemma 95,2; UPOS 93,7	—
UD validatsiyasi	✔ o‘tgan	✘ 92 ogohl.
Noproyektiv daraxtlar	11,0 %	6,8 %
Bog‘lanish turlari	38	33 (jami 39)

Tajriba rejasi: 3×2 faktorli dizayn

Vektor / birlashtirish	UzUDT 435 / 48 / 198 gap	UzUDT+UT† 781 / 78 / 325 gap
FastText, 300	E1.1	E1.2
BERT + oxirgi	E2.1	E2.2
BERT + o‘rtacha	E3.1	E3.2‡

† bu sharoitda parserga oltin UPOS/XPOS/UFeats berilgan. ‡ E3.2 jurnali repozitoriyda yo‘q.

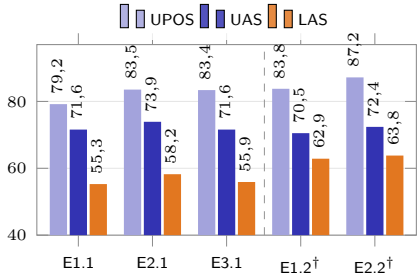


1 000 tokenga to‘g‘ri keladigan soni

Korpuslar bir-birini to‘ldiradi. Manba: data/udbase/*.conllu.

Natijalar: validatsiya to‘plami, eng yaxshi nazorat nuqtasi

		Teglash		Sintaktik tahlil	
Tajriba	**Vektorlar**	UPOS	UFeats	UAS	LAS
A. UzUDT; parserga bashorat qilingan teglar (48 gap, 644 token)					
E1.1	FastText	79,19	66,61	71,58	55,28
E2.1	BERT + oxirgi	**83,54**	**68,17**	**73,91**	**58,23**
E3.1	BERT + o‘rtacha	83,39	65,84	71,58	55,90
B. UzUDT+UT; parserga oltin teglar† *(78 gap, 851 token)*					
E1.2	FastText	83,78	70,98	70,51	62,87
E2.2	BERT + oxirgi	**87,19**	**73,21**	**72,39**	**63,81**



Xulosalar

1. Sintaktik tahlilda «oxirgi qism so‘z» operatori o‘rtacha qiymatdan **ishonchli** yaxshi.

2. TahrirchiBERT teglashni yaxshilaydi: UPOS $+3{,}4 \ldots +4{,}4$ punkt.

3. Oltin morfologiya berilganda BERT ning sintaktik ustunligi ahamiyatsiz.

Juftlashgan bootstrap, $B = 10^4$

$\Delta_b^* = \mathrm{LAS}_A(S_b^*) - \mathrm{LAS}_B(S_b^*)$; 95 % IO $= \left[\Delta^*_{(0{,}025)}; \Delta^*_{(0{,}975)}\right]$

oxirgi – o‘rt.: $+2{,}64$ $[+0{,}97; +4{,}12]$, $p \approx 0{,}002$ ✓

BERT – FT (A): $+2{,}95$ $[0{,}00; +5{,}63]$, $p \approx 0{,}03$

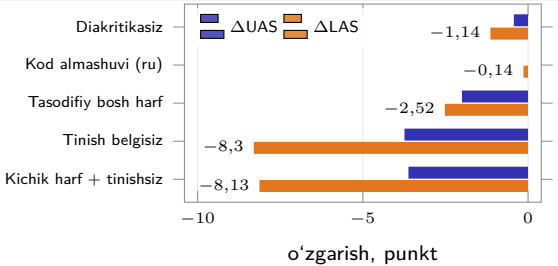
BERT – FT (B): $-1{,}29$ $[-4{,}66; +2{,}23]$ — ahamiyatsiz

† A va B bevosita taqqoslanmaydi: kiruvchi teglar (bashorat / oltin) va validatsiya to‘plamlari farqli. Bootstrap — oxirgi qadam bashoratlari bo‘yicha.

Test to‘plamida baholash: turg‘unlik va dekodlash usullarini taqqoslash

Ta’rif 6 (sezgirlik)

π — matnni buzuvchi (g‘alayon) operatori, ρ_π — o‘zgargan tokenlar ulushi; sezgirlik: $S_\pi = \mathrm{LAS}(\mathcal{D}) - \mathrm{LAS}\big(\pi(\mathcal{D})\big)$.



E2.1 modeli, UzUDT test to‘plami (198 gap, 2 101 token); toza matnda UAS 71,06, LAS 54,07. ρ_π: diakritika 8,7 %, kod almashuvi 0,2 %, bosh harf 53,3 %, tinish belgisi 19,6 %.

MST va ochko‘z dekodlash (bir xil test)

	Graf + MST Stanza, E2.1	O‘tish (arc-eager) spaCy, S1.1	Δ
UAS	**71,06**	67,72	+3,34
LAS	**54,07**	45,35	+8,72

Ikkalasi ham TahrirchiBERT; spaCy baholovchisi tinish belgilarini hisobga olmaydi.

Talqin

1. Global (MST) optimallashtirish ochko‘z ketma-ket qarorlardan **8,7 punkt** yaxshi: xatolar zanjir bo‘ylab tarqalmaydi.
2. Diakritika yo‘qolishiga turg‘un ($-1{,}1$), kod almashuviga deyarli befarq ($-0{,}1$).
3. Tinish belgilari — muhim tuzilmaviy signal: ularsiz LAS $8{,}3$ punktga tushadi.

Dasturiy majmua

Ma'lumotlar

UzUDT (UD v2.17, CC BY-SA 4.0) va UT

CoNLL-U o'quvchi va birlashtiruvchi

spaCy DocBin konvertori

korpus statistikasi

O'qitish

Stanza: BERT kanali va Π operatori (`--bert_model`, `--bert_pooling`)

spaCy uchun uz til moduli

CSV / W&B jurnallari

Baholash

CoNLL 2018 metrikalari (UAS, LAS, MLAS, BLEX)

juftlashgan bootstrap

g'alayon (turg'unlik) testlari

MST vaqt tajribasi

E'lon qilish

HuggingFace: `Sanatbek/uzudt`

GitHub: `robust-parsing-uzbek`

UD rasmiy repozitoriysi

</> Men Stanza kutubxonasiga kiritgan o'zgarishlar

`depparse/model.py` — tahlilchiga BERT kanali, ildiz uchun $\mathbf{e}_0 = \mathbf{0}$

`common/bert_embedding.py` — Π operatori: `last` / `mean`

`depparse/data.py`, `trainer.py` — xom tokenlar, BERT ni ulash

`pos/model.py`, `tagger.py`, `parser.py` — teglovchi, CLI, jurnallar

`common/constant.py` — uz (o'zbek tili) kodi

Hajmi va texnik ko'rsatkichlar

Stanza: ≈570 qator (20 fayl); spaCy-uz moduli: ≈920; baholash skriptlari: ≈1 600 ⇒ **≈3 100 qator** Python kodi.

Python, PyTorch, Stanza, spaCy 3.8; o'qitish — NVIDIA RTX A6000.

Xulosa

1. O‘zbek tilining badiiy va ta’limiy matnlari uchun **ilk bor** oltin standart UD korpusi **UzUDT** (684 gap, 7 582 token) yaratildi va e’lon qilindi.
2. Sintaktik tahlil **graflar nazariyasi** tilida qo‘yildi: gap — vaznli to‘liq yo‘naltirilgan graf, tahlil natijasi — maksimal vaznli qamrovchi daraxt; daraxtlar soni $n^{\,n-1}$.
3. Stanza bi-affin tahlilchisi o‘zgartirildi: TahrirchiBERT kanali va Π operatori ($\mathbb{R}^{500} \to \mathbb{R}^{1143}$); **ilk bor** «oxirgi qism so‘z» operatorining ustunligi tasdiqlandi ($+2{,}6$ LAS, $p \approx 0{,}002$).
4. Daraxtni izlash murakkabligi $n^{\,n-1}$ dan $O(n^2)$ ga tushirildi: 11 so‘zli gapda $\approx 1{,}7$ sutka $\to$ 0,42 ms; MST ochko‘z o‘tish usulidan LAS bo‘yicha **8,7 punkt** yuqori.
5. Adam$\to$AMSGrad bilan o‘qitishda 1 000 qadamda (6,3 daqiqa) LAS $24{,}7 \to 58{,}2\,\%$; model diakritikaga turg‘un, tinish belgilariga sezgir ($-8{,}3$ LAS).
6. **Dasturiy majmua** ($\approx$3 100 qator Python) yaratildi, korpus va modellar ochiq e’lon qilindi.

Kelgusi ishlar

Barcha tajribalarni yagona bo‘linma va bashorat qilingan teglar bilan **test to‘plamida** qayta baholash; ko‘p urug‘li (seed) tajribalar; global normallash (Kirchhoff matritsa-daraxt teoremasi, $Z(x) = \det \widehat{\mathbf{L}}$); BERTbek va FastText+BERT birikmasi.

684 gap	$n^{\,n-1} \to O(n^2)$	$+2{,}6$ **LAS**	**58,2 LAS**
UzUDT korpusi	daraxtni izlash (vaqt)	oxirgi qism so‘z ($p \approx 0{,}002$)	1 000 qadamda (Adam)

Natijalarning aprobatsiyasi va e'lon qilinganligi

1. Matlatipov, S., Aripov, M., Bobokandov, M., & Matlatipov, G. (2026). Annotated Universal Dependencies dataset for literary and educational Uzbek texts. *Data in Brief*, *66*, Article 112857. https://doi.org/10.1016/j.dib.2026.112857 **[Scopus, Q1]**
2. Matlatipov, S. (2027). Towards robust Uzbek neural dependency parsing: Cross-treebank training. In E. Cabrio & E. Monteiro (Eds.), *Natural language processing and information systems* (pp. 204–218). Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-29532-3_15 **[Scopus, Q2]**
3. Matlatipov, S., Ismoilova, M., & Aripov, M. (2026). Graph-based vs transition-based dependency parsing for Uzbek. In *2026 IEEE 27th International Conference of Young Professionals in Electron Devices and Materials (EDM)* (pp. 1–6). IEEE. https://doi.org/10.1109/EDM69524.2026.11632281 **[Scopus, Q3]**
4. Matlatipov, S. G., & Aripov, M. (2026). UzUDT: Uzbek Universal Dependencies treebank. In S. Piperidis, N. Bel, H. van den Heuvel, N. Ide, S. Krek, & A. Toral (Eds.), *Proceedings of the Fifteenth Language Resources and Evaluation Conference* (pp. 11642–11649). ELRA Language Resource Association. https://doi.org/10.63317/2uedjqezjxn5 **[Scopus]**
5. Matlatipov, S., & Kurbanova, M. (2026). A gold-standard dependency treebank for the Uzbek language. In *2026 IEEE Ural-Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology (USBEREIT)* (pp. 1–3). IEEE. https://doi.org/10.1109/USBEREIT70063.2026.11580531 **[Scopus]**
6. Matlatipov, S. (2026). Neural morphosyntactic tagging and dependency parsing for Uzbek. In *2026 IEEE Ural-Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology (USBEREIT)* (pp. 1–4). IEEE. https://doi.org/10.1109/USBEREIT70063.2026.11580890 **[Scopus]**

E’tiboringiz uchun rahmat!

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Zaxira slayd: kontekstni hisobga oluvchi kodlovchi — highway BiLSTM

Kirish vektori

$$\mathbf{v}_i = \left[\mathbf{e}_i^{w};\ \mathbf{e}_i^{\text{lem}};\ \mathbf{e}_i^{\text{upos}} + \mathbf{e}_i^{\text{xpos}};\ \sum\nolimits_k \mathbf{e}_i^{f_k};\ \mathbf{W}_c\mathbf{c}_i;\ \mathbf{e}_i\right] \in \mathbb{R}^{1143}$$

so'z 75 + lemma 75 + teglar 50 + belgilar 50 + harf-LSTM 125 + BERT 768

(FastText variantida $\mathbf{e}_i$ o'rniga $\mathbf{W}_p\mathbf{p}_i \in \mathbb{R}^{125}$, jami 500)

LSTM yacheykasi (S. Hochreiter, J. Schmidhuber, 1997)

$$\begin{pmatrix}\mathbf{i}_t\\ \mathbf{f}_t\\ \mathbf{o}_t\\ \mathbf{g}_t\end{pmatrix} = \begin{pmatrix}\sigma\\ \sigma\\ \sigma\\ \tanh\end{pmatrix}\left(\mathbf{W}\begin{bmatrix}\mathbf{v}_t\\ \mathbf{h}_{t-1}\end{bmatrix} + \mathbf{b}\right),$$

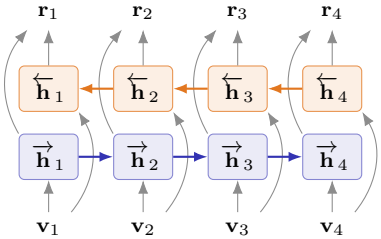
$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t,$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t).$$

Ikki yo'nalish va highway bog'lanish, $\ell = 1, 2, 3$

$$\mathbf{z}_t^{(\ell)} = \left[\overrightarrow{\mathbf{h}}_t^{(\ell)}; \overleftarrow{\mathbf{h}}_t^{(\ell)}\right] + \sigma\left(\mathbf{G}^{(\ell)}\mathbf{z}_t^{(\ell-1)}\right) \odot \tanh\left(\mathbf{W}_H^{(\ell)}\mathbf{z}_t^{(\ell-1)}\right),$$

$$\mathbf{r}_i = \mathbf{z}_i^{(3)} \in \mathbb{R}^{800}$$



Gradient oqimi

To'g'ridan-to'g'ri yo'l bo'yicha $\dfrac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} = \operatorname{diag}(\mathbf{f}_t)$: unutish darvozasi uzoq bog'lanishlarda gradientning so'nishini yumshatadi.

Teglovchi: 2 qatlam, 2×200; parser: 3 qatlam, 2×400.